

P8106 Final Project - Predicting COVID-19 Severity

Part I: Exploratory Analysis

A dataset of 1,000 participants was randomly sampled from the full dataset of 10,000 using a fixed random seed (3352) to ensure reproducibility, which is also the seed used for all randomization throughout this analysis. The dataset was then split into a training set (80%, $n = 800$) and a test set (20%, $n = 200$) using stratified sampling to preserve the outcome class distribution across both sets. There were no missing values in the sampled dataset.

The outcome variable, COVID-19 severity, was moderately imbalanced, with 32% of training participants ($n = 256$) classified as severe and 68% ($n = 544$) as not severe.

Table 1 presents descriptive statistics for all variables stratified by severity status in the training set. Among continuous predictors, severe cases were older on average (61.4 vs. 59.6 years), had higher BMI (28.3 vs. 27.5), higher systolic blood pressure (133.0 vs. 128.7 mmHg), and higher LDL cholesterol (112.4 vs. 107.5 mg/dL). Among categorical predictors, vaccination status showed the most striking separation: 81% of severe cases were unvaccinated compared to only 21% of non-severe cases, suggesting a strong protective effect of vaccination. Hypertension was more prevalent among severe cases (60% vs. 43%), as was diabetes (21% vs. 13%). Gender, race, and smoking status showed little difference in severity proportions across categories.

Figure 1 displays boxplots of continuous predictors stratified by severity group. Visible separation between severe and not-severe groups is apparent for SBP, age, and BMI, while depression, weight, and height show minimal visual separation, consistent with the patterns observed in Table 1. Figure 2 displays proportional bar charts for all categorical predictors. The vaccine panel shows the highest contrast across all categorical variables, with unvaccinated individuals comprising the large majority of severe cases. Hypertension and diabetes panels also show visible differences in severity proportions, while gender, race, and smoking status appear largely uniform across their respective category levels.

Part II: Model Training

Four classification models were trained on the training set to predict COVID-19 severity: logistic regression (GLM), penalized logistic regression with elastic net regularization (GLMN), gradient boosting with AdaBoost (GBM), and support vector machine with a radial kernel (SVM). GLM and GLMN serve as the simpler comparator models. GBM and SVM represent the advanced methods of primary interest to the researchers. By fitting all four models under identical cross-validation conditions, the

analysis directly addresses whether the added complexity of boosting and SVM translates into meaningful gains in predictive performance over the simpler alternatives.

All models were trained using 10-fold cross-validation with area under the ROC curve (AUC) as the performance metric, implemented through the caret package. AUC was selected over accuracy because the outcome is moderately imbalanced (32% severe); accuracy would be misleading as a naive classifier predicting “not severe” for all observations would achieve 68% accuracy. AUC evaluates model discrimination across all possible classification thresholds and is not affected by class imbalance.

Logistic Regression (GLM)

The logistic regression model requires no tuning parameters and was fitted directly within the 10-fold cross-validation framework using method = "glm" in caret. Despite having no hyperparameters to optimize, fitting GLM through the same caret train() function and trainControl object as the other three models ensures that its cross-validated AUC is evaluated on identical folds, making the subsequent resamples() comparison valid and apples-to-apples across all four models. The model estimates the log-odds of severe COVID-19 as a linear combination of all 13 predictors via logistic transformation.

Penalized Logistic Regression with Elastic Net Regularization (GLMN)

The elastic net model was tuned over a grid of mixing parameters (alpha: 0, 0.2, 0.4, 0.6, 0.8, 1.0) and regularization strengths (lambda: exp(seq(-12, -1, length = 20))). The optimal parameters were alpha = 0.4 and lambda = 0.000198, corresponding to an elastic net blend with light regularization. The tuning plot (Figure 3) demonstrates that the optimal parameter combination is interior to the grid.

Boosting with AdaBoost (GBM)

The AdaBoost model was tuned over a grid of boosting iterations (n.trees: 100, 200, 500, 1000, 2000, 3000, 5000), tree depth (interaction.depth: 1, 2, 3), and learning rate (shrinkage: 0.001, 0.003, 0.005), with minimum node size fixed at 10 following the lecture code convention. The optimal configuration was n.trees = 1000, interaction.depth = 2, and shrinkage = 0.005. The tuning plot (Figure 4) shows the full pattern across tree counts for each shrinkage and depth combination, confirming the optimal point is interior to the grid.

Support Vector Machine with a Radial Kernel (SVM)

The SVM model used a radial kernel and was tuned over a grid of cost parameters (C: exp(seq(6, 12, length = 15))) and kernel bandwidth parameters (sigma: exp(seq(-14, -8, length = 6))). Unlike the three previous models, SVM does not naturally produce predicted probabilities. For this reason, a separate trainControl object was used for SVM with default

accuracy and kappa as the evaluation metrics, consistent with the lecture approach for SVM tuning. The optimal parameters were $C = 44994$ and $\sigma = 3.04e-05$, both interior to the tuning grid (Figure 5). The large optimal C value is consistent with data that is nearly linearly separable in the kernel-transformed feature space.

Part III: Model Comparison

Since SVM was tuned on accuracy while GLM, GLMN, and GBM were tuned on ROC, a unified four-model cross-validation comparison requires a common metric. GLM, GLMN, and GBM were therefore refitted using the same trainControl as SVM, a 10-fold cross-validation with default accuracy and kappa, allowing all four models to enter a single resamples() comparison.

Cross-validated performance was comparable across all four models (Table 2), with mean accuracy ranging from 0.836 (GLM) to 0.851 (GLMN and GBM) and overlapping interquartile ranges across all 10 folds (Figure 6). The differences in accuracy are small, with less than two percentage points separating the best and worst performing models, indicating that no model meaningfully outperformed the others.

Among the four models, GLMN achieved the highest mean accuracy (0.851) with the narrowest interquartile range, indicating superior consistency across folds relative to other models. Though GLMN and GBM had almost identical mean and median accuracy, since GLMN is very interpretable and computationally economic, GLMN was therefore selected as the final model.

Part IV: Results

Question 1: Do Boosting and SVM Outperform Simpler Models?

Model performance was evaluated on the held-out test set of 200 participants using confusion matrix metrics, enabling a direct four-model comparison including SVM (Table 3). Test set accuracy ranged from 0.825 (GLM) to 0.840 (GBM), a difference of only 0.015. GLMN and SVM achieved identical accuracy (0.830) on the test set, and GBM's only higher by 0.01 over GLMN. These results are consistent with the cross-validation findings and confirm that the advanced methods do not provide meaningful predictive advantages over simpler, interpretable models in this dataset.

ROC curves were constructed for GLM, GLMN, and GBM on the test set (Figure 7). SVM was excluded from the ROC comparison as it does not naturally produce predicted probabilities. The three curves overlap substantially across the full range of thresholds, with AUC values within 0.003 of each other, visually confirming the equivalence of discriminative performance across the probabilistic models.

Taken together, the cross-validation and test set results show that boosting and SVM do not perform better than the simpler models for this dataset. All four models have similar performance, and the more complex methods do not provide clear improvement in prediction.

Question 2: Predictive Risk Score

Individual predicted probabilities of severe COVID-19 infection(predictive risk score) can be generated using the final selected model GLMN for each of the 200 test set participants. For each individual in the test set, the model produces a predicted probability between 0 and 1 via the logistic transformation applied to a linear combination of all 13 predictors. Individuals with predicted probability above 0.5 are classified as severe; those at or below 0.5 are classified as not severe.

The model demonstrated strong discrimination at both extremes of risk (Table 4). At the standard 0.5 classification threshold, GLMN achieved accuracy of 0.830, sensitivity of 0.734, specificity of 0.875, and kappa of 0.609, with equal false positive and false negative counts (17 each) indicating a well-balanced classifier.

Question 3: Key Predictors and Their Influence

Variable importance was assessed using absolute standardized coefficients from GLMN and relative influence from GBM, where the percentage reduction in loss attributable to each predictor across all boosting iterations, providing complementary linear and nonlinear perspectives (Figures 8 and 9). Vaccination status was the dominant predictor in both models, accounting for 49.6% of GBM's total relative influence and scoring approximately twice the importance of the second-ranked predictor in GLMN. BMI ranked among the top three in both models, while SBP ranked second in GBM but lower in GLMN, which likely reflects collinearity with hypertension, with each model selecting a different representative of the same blood pressure signal. Smoking, gender, and race showed near-zero importance in both models, consistent with EDA findings.

Partial dependence plots from GBM for the four most influential predictors reveal the shape of these relationships (Figure 10). Vaccination status showed the largest marginal effect as unvaccinated individuals had an average predicted severity probability of 0.61 versus 0.13 for vaccinated individuals, a difference of 48 percentage points. SBP exhibited a threshold effect, remaining flat below 122 mmHg before rising sharply between 122 and 140 mmHg and plateauing around 0.45. BMI showed a similar inflection between 27 and 33, plateauing at 0.55. Age showed a more gradual but accelerating increase in predicted severity probability, with risk rising modestly through middle age and more steeply above age 65.

Appendix

Table 1. Descriptive statistics for all variables stratified by COVID-19 severity status (training set, n = 800). Continuous variables are summarized as mean (SD); categorical variables as n (%).

Characteristic	Overall N = 800¹	severe N = 256¹	not_severe N = 544¹
age	60.2 (4.4)	61.4 (4.3)	59.6 (4.4)
gender			
Female	418 (52%)	140 (55%)	278 (51%)
Male	382 (48%)	116 (45%)	266 (49%)
race			
White	509 (64%)	157 (61%)	352 (65%)
Asian	51 (6.4%)	16 (6.3%)	35 (6.4%)
Black	163 (20%)	59 (23%)	104 (19%)
Hispanic	77 (9.6%)	24 (9.4%)	53 (9.7%)
smoking			
Never	494 (62%)	148 (58%)	346 (64%)
Former	234 (29%)	84 (33%)	150 (28%)
Current	72 (9.0%)	24 (9.4%)	48 (8.8%)
height	170.2 (5.8)	169.2 (5.9)	170.6 (5.8)
weight	80.1 (6.9)	80.7 (6.8)	79.8 (7.0)
bmi	27.7 (2.7)	28.3 (3.0)	27.5 (2.5)
diabetes	126 (16%)	53 (21%)	73 (13%)
hypertension	385 (48%)	153 (60%)	232 (43%)
SBP	130.1 (8.1)	133.0 (7.2)	128.7 (8.2)
LDL	109.0 (19.6)	112.4 (18.8)	107.5 (19.8)
vaccine			
Unvaccinated	323 (40%)	207 (81%)	116 (21%)
Vaccinated	477 (60%)	49 (19%)	428 (79%)
depression	7.0 (2.0)	6.9 (2.0)	7.1 (2.0)
¹ Mean (SD); n (%)			

Table 2. Cross-validated performance comparison across all four models using 10-fold cross-validation with accuracy and kappa as evaluation metrics.

Accuracy							
Model	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
GLM	0.8000	0.8148	0.8261	0.8365	0.8455	0.9114	0
GLMN	0.8000	0.8406	0.8554	0.8513	0.8633	0.8987	0
GBM	0.8000	0.8255	0.8554	0.8514	0.8707	0.9012	0
SVM	0.7901	0.8276	0.8500	0.8451	0.8611	0.8987	0
Kappa							
Model	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
GLM	0.5442	0.5723	0.6052	0.6212	0.6374	0.7884	0
GLMN	0.5325	0.6260	0.6628	0.6516	0.6987	0.7608	0
GBM	0.5442	0.5974	0.6519	0.6507	0.6880	0.7734	0
SVM	0.4817	0.6049	0.6546	0.6423	0.6855	0.7659	0

Table 3. Test set performance summary for all four models evaluated on the held-out test set (n = 200).

Model	Accuracy	Sensitivity	Specificity	Kappa
GLM	0.825	0.750	0.860	0.603
GLMN	0.830	0.734	0.875	0.609
GBM	0.840	0.734	0.890	0.629
SVM	0.830	0.734	0.875	0.609

Table 4. Predicted risk scores for the ten highest-risk and ten lowest-risk individuals in the test set, generated from the final GLMN model. Prob_Severe denotes the predicted probability of severe COVID-19 infection.

Highest Risk			
ID	Actual	Prob_Severe	Predicted
60	severe	0.999	severe
101	severe	0.991	severe
89	severe	0.982	severe
148	severe	0.962	severe
157	severe	0.954	severe
37	severe	0.952	severe
65	severe	0.939	severe
66	severe	0.935	severe
142	severe	0.932	severe
107	severe	0.923	severe
Lowest Risk			
ID	Actual	Prob_Severe	Predicted
56	not_severe	0.012	not_severe

64	not_severe	0.011	not_severe
15	not_severe	0.009	not_severe
99	not_severe	0.005	not_severe
117	not_severe	0.005	not_severe
11	not_severe	0.004	not_severe
59	not_severe	0.004	not_severe
166	not_severe	0.004	not_severe
141	not_severe	0.003	not_severe
28	not_severe	0.002	not_severe

Figure 1. Boxplots of continuous predictors stratified by COVID-19 severity status (severe vs. not severe) in the training set.

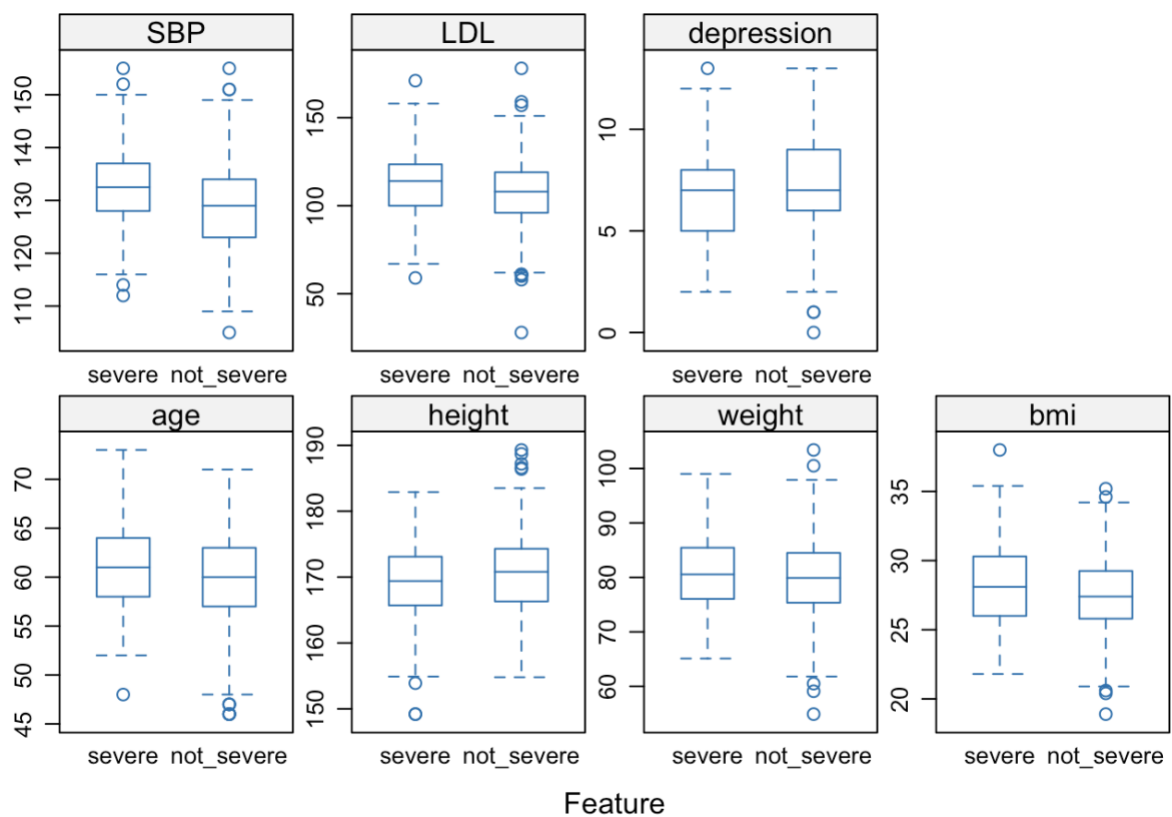


Figure 2. Proportional bar charts of categorical predictors stratified by COVID-19 severity status in the training set.

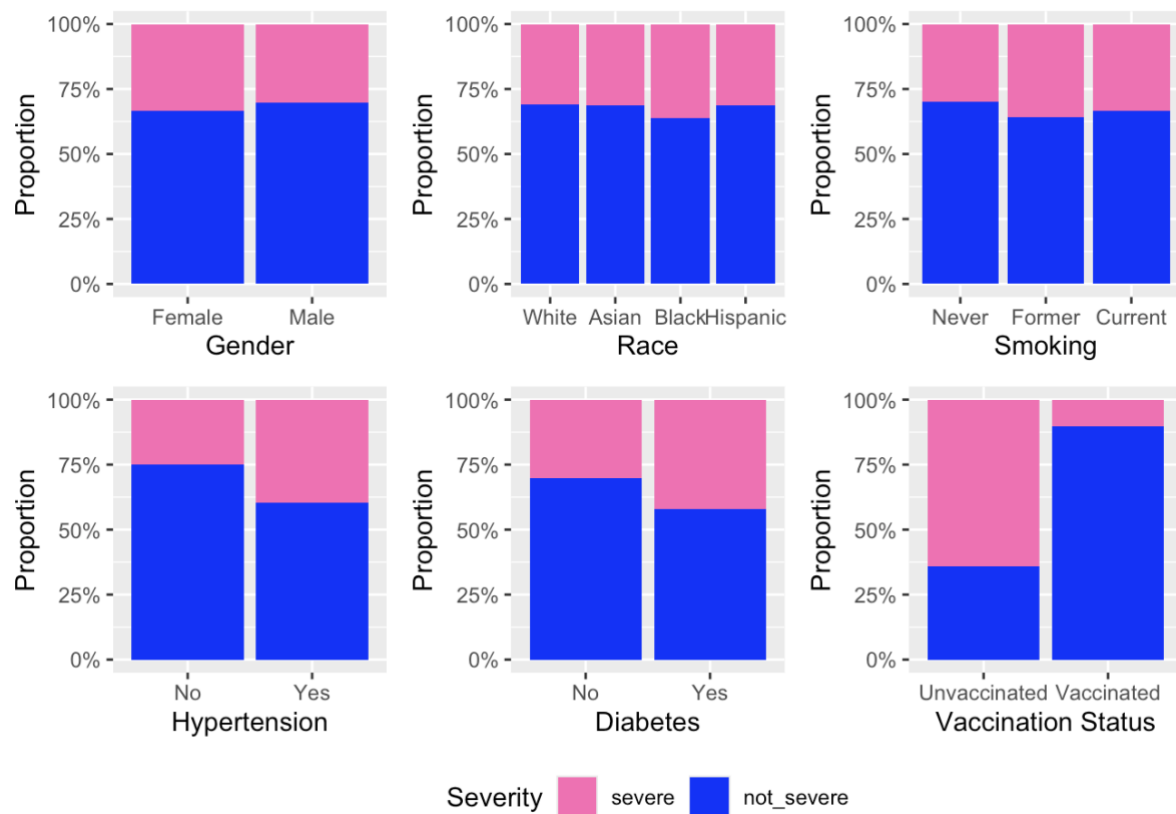


Figure 3. Cross-validation tuning plot for the elastic net model. Each line represents a mixing parameter value (alpha), and the x-axis shows the log-transformed regularization parameter (lambda).

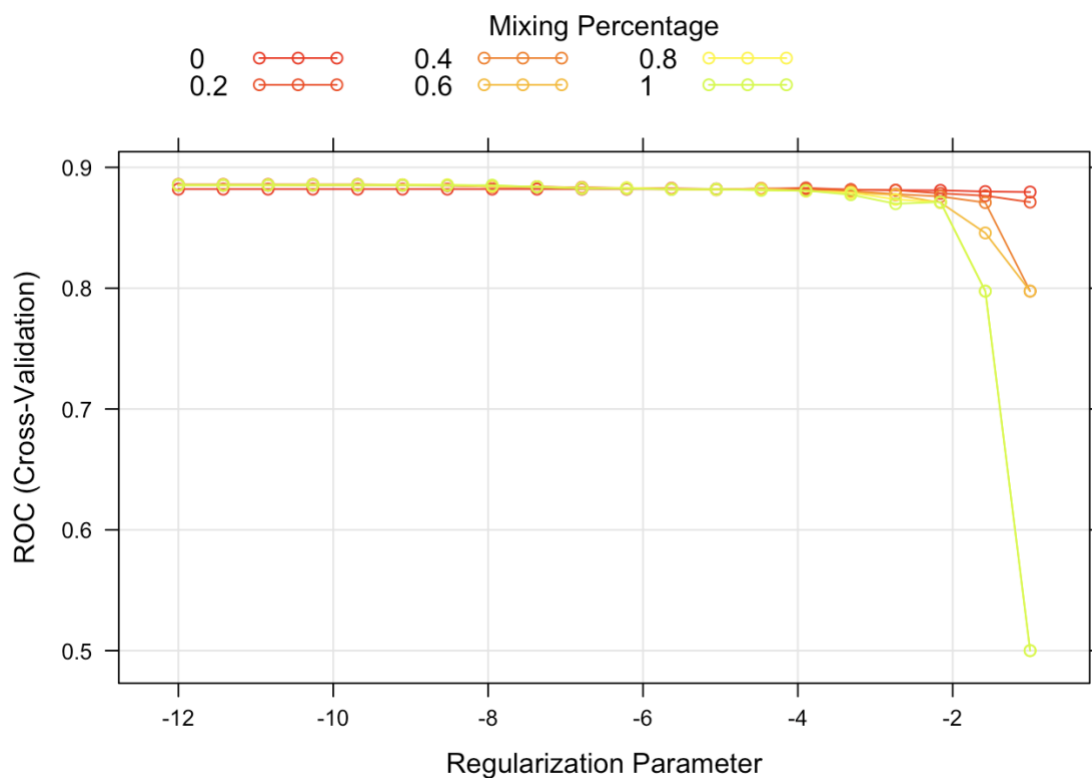


Figure 4. Cross-validation tuning plot for the AdaBoost model. Panels correspond to learning rate (shrinkage) values. Lines represent maximum tree depth (interaction.depth). The x-axis shows number of boosting iterations.

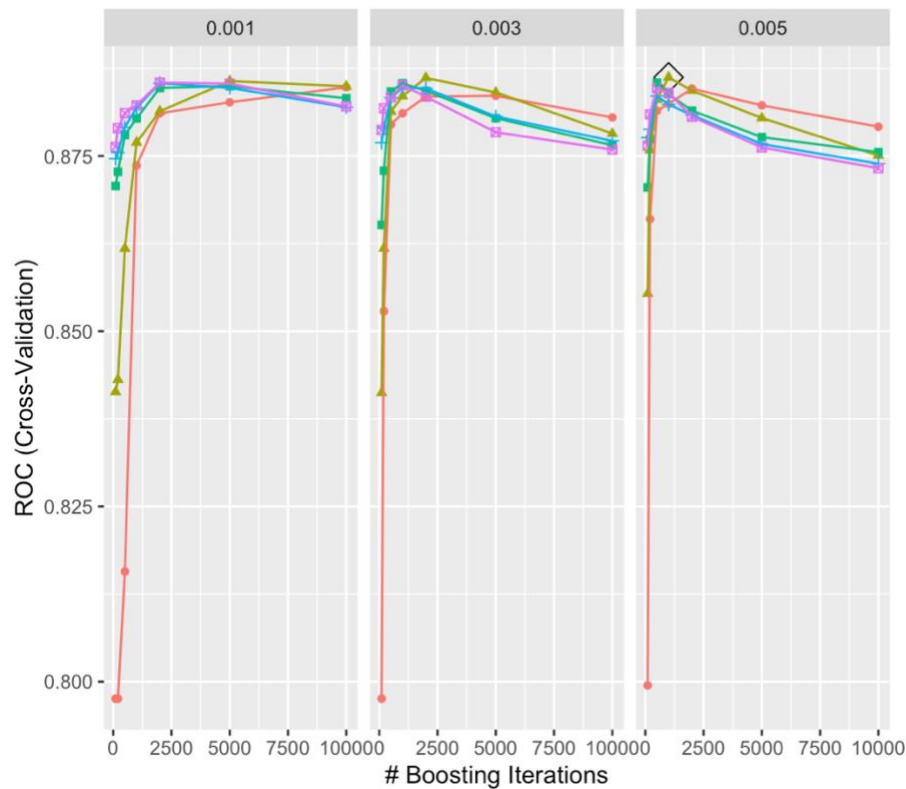


Figure 5. Cross-validation tuning plot for the SVM radial kernel model. Lines represent sigma (kernel bandwidth) values and the x-axis shows the cost parameter on a log scale.

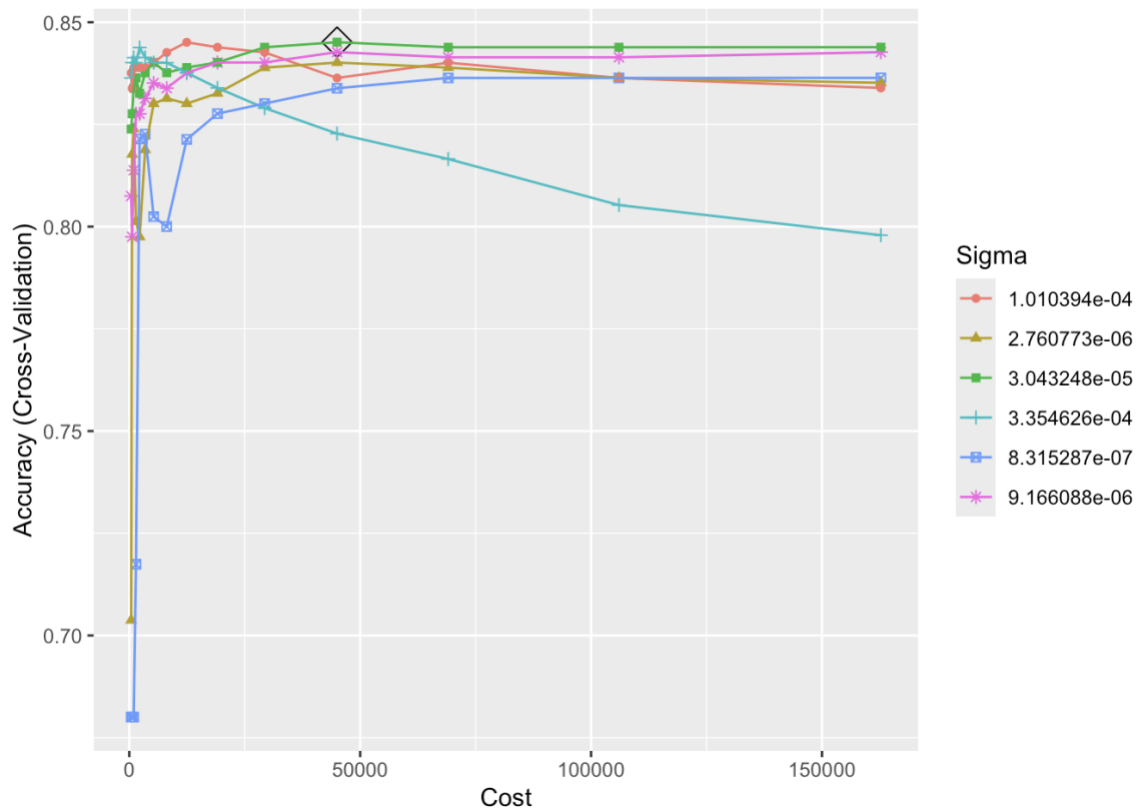


Figure 6. Box-and-whisker plots comparing cross-validated accuracy and kappa across all four models from 10-fold cross-validation.

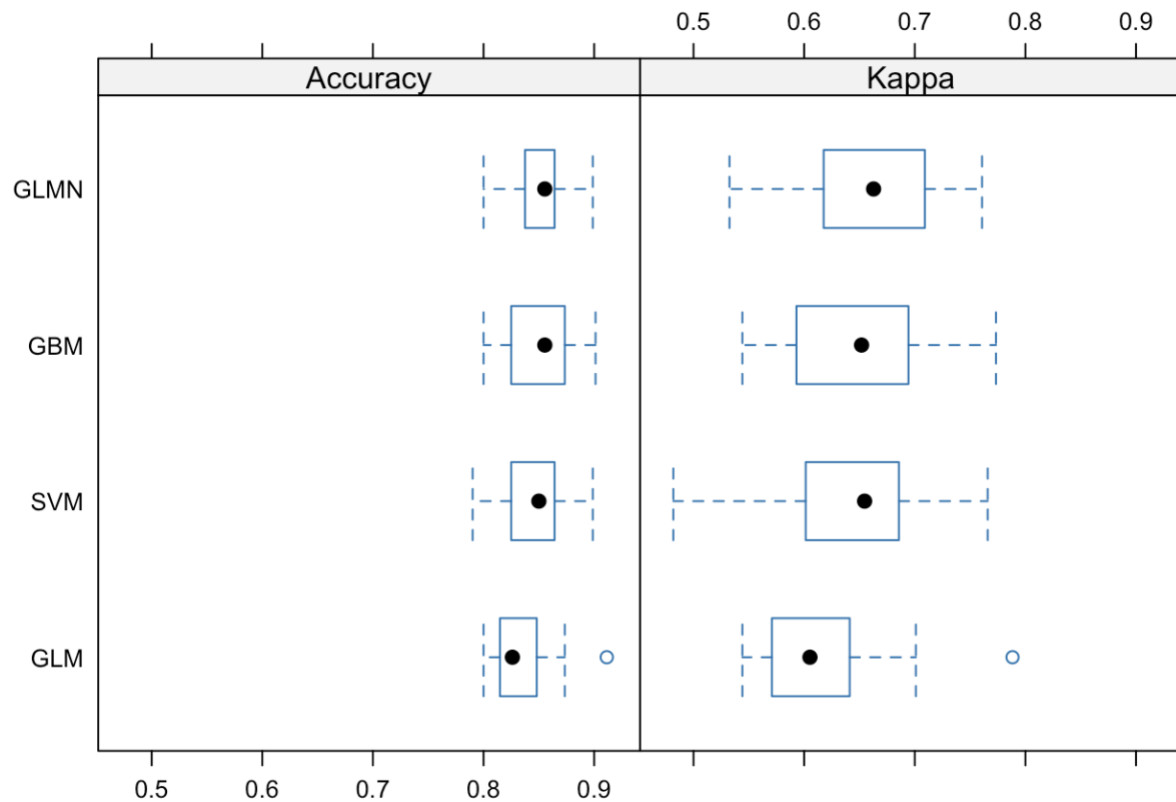


Figure 7. ROC curves for GLM, GLMN, and GBM evaluated on the held-out test set (n = 200). AUC values are reported in the legend.

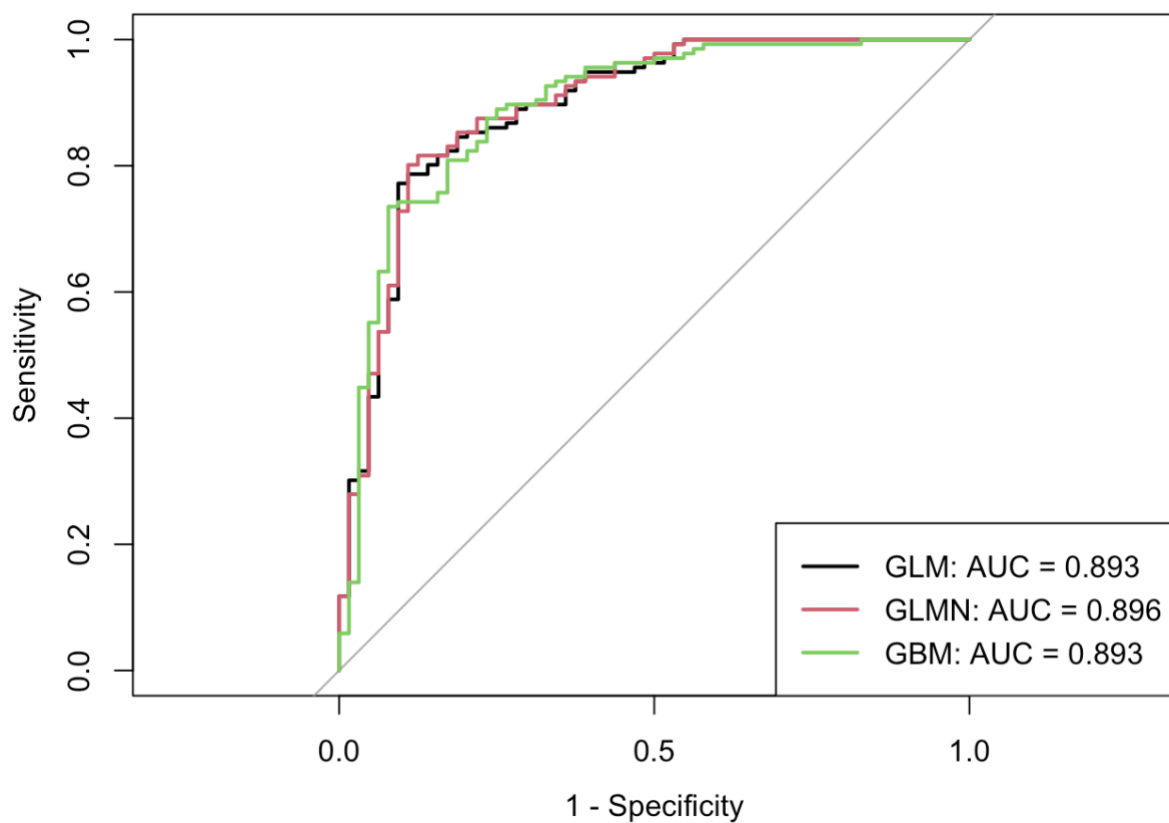


Figure 8. Variable importance from the final selected model (GLMN) based on absolute standardized coefficients after elastic net regularization.

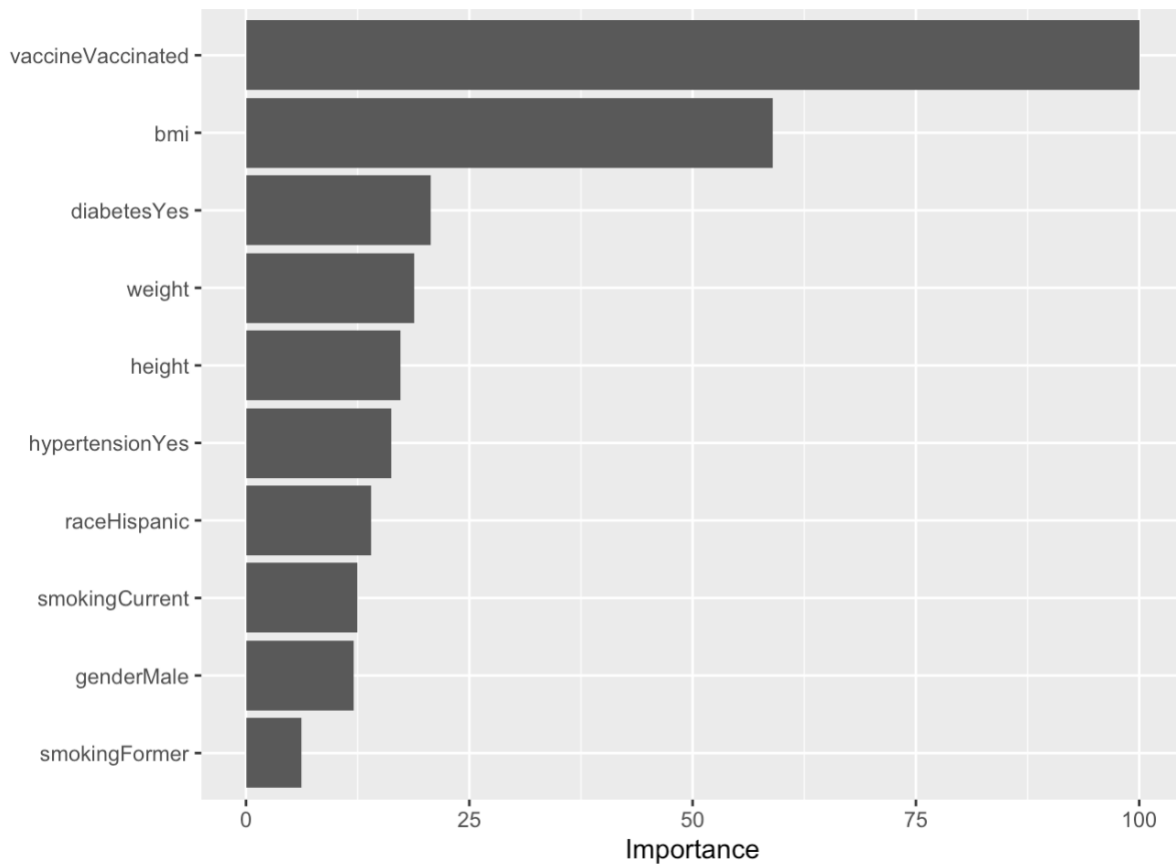


Figure 9. Variable importance from the AdaBoost model based on relative influence.

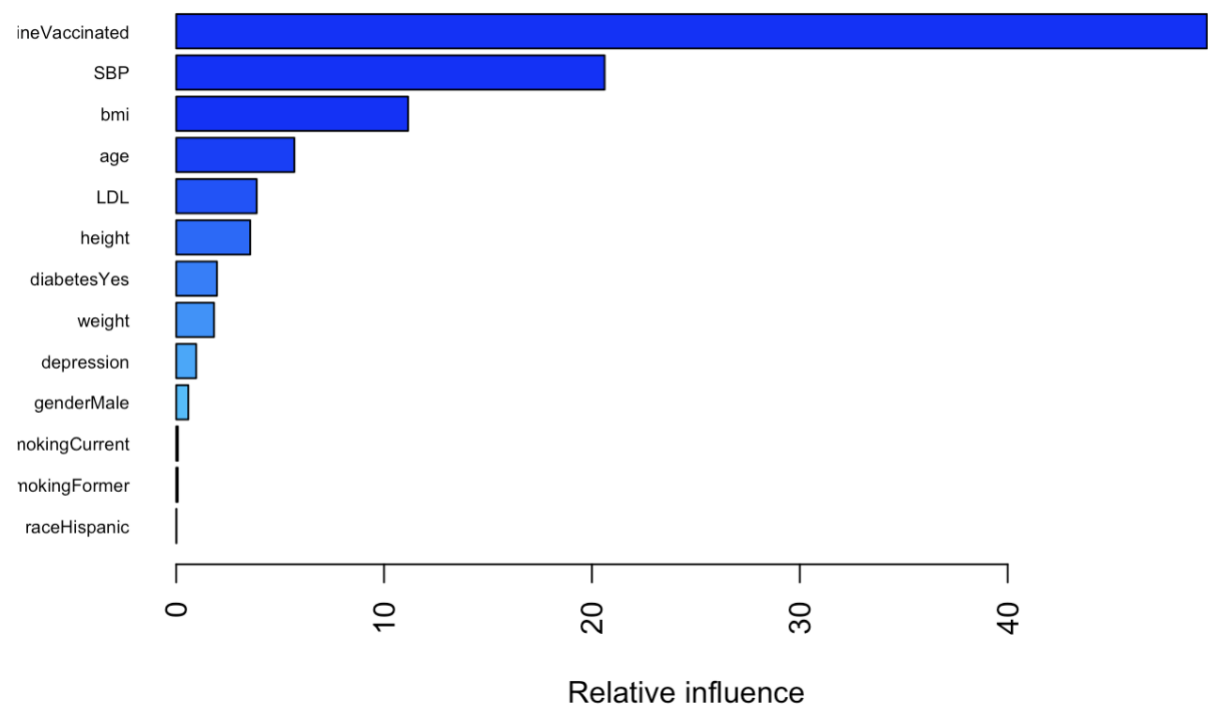


Figure 10. Partial dependence plots for the four most influential predictors. The y-axis (yhat) represents the average predicted probability of severe COVID-19 infection.

